# Driving Project Success with Real-Time Monitoring

Maximizing Impact Through Effective Progress Tracking

Patel Rohan

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# Agenda

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2. System Overview and Goals
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# Introduction and Challenges

# Why Project Monitoring Matters

### Keeps Projects Aligned with Goals

- Monitors progress against **time, cost, and scope** baselines

### Enables Proactive Risk Management

- Identifies **delays, risks, and bottlenecks** early for timely intervention

### Empowers Data-Driven Decisions

- Provides real-time dashboards and reports for **informed management actions**

# Challenges in Traditional Tracking

## Inefficiency of Manual Tracking

- Spreadsheets are cumbersome for **distributed teams**
- Manual updates increase risk of **errors and omissions**

## Delayed Visibility and Response

- Lack of **real-time updates** hinders early intervention

## Fragmented Project Data

- Difficult to consolidate information for **timely reporting**

# System Overview and Goals

# System Overview at a Glance

### Unified Project Management Platform

- Centralizes all project data for real-time visibility
- Supports both web and mobile access for all user roles

### Modern, Scalable Architecture

- Built with **React** for web and **React Native** for mobile
- Backend powered by **ASP.NET Core** using Clean Architecture principles

### Secure and Connected Ecosystem

- RESTful APIs ensure seamless integration between clients and backend
- Role-based access and encrypted communication for data protection

# Key Goals of the Solution

### Instant Project Health Insights

- Centralized dashboards provide **real-time status** on schedule, cost, and scope.
- Visual indicators highlight **delays, risks, and bottlenecks** for immediate attention.

### Proactive Management & Control

- Automated alerts and notifications enable **timely corrective actions**.
- Historical progress tracking supports **trend analysis** and decision-making.

### Seamless Team Collaboration

- Web and mobile apps ensure **continuous updates** from distributed teams.
- Role-based access keeps data secure and relevant for all stakeholders.

# Who Benefits from the System

## Empowering Project Teams

- Designed for IT, construction, and government sectors
- Streamlines collaboration for distributed teams

## Supporting Managers & Coordinators

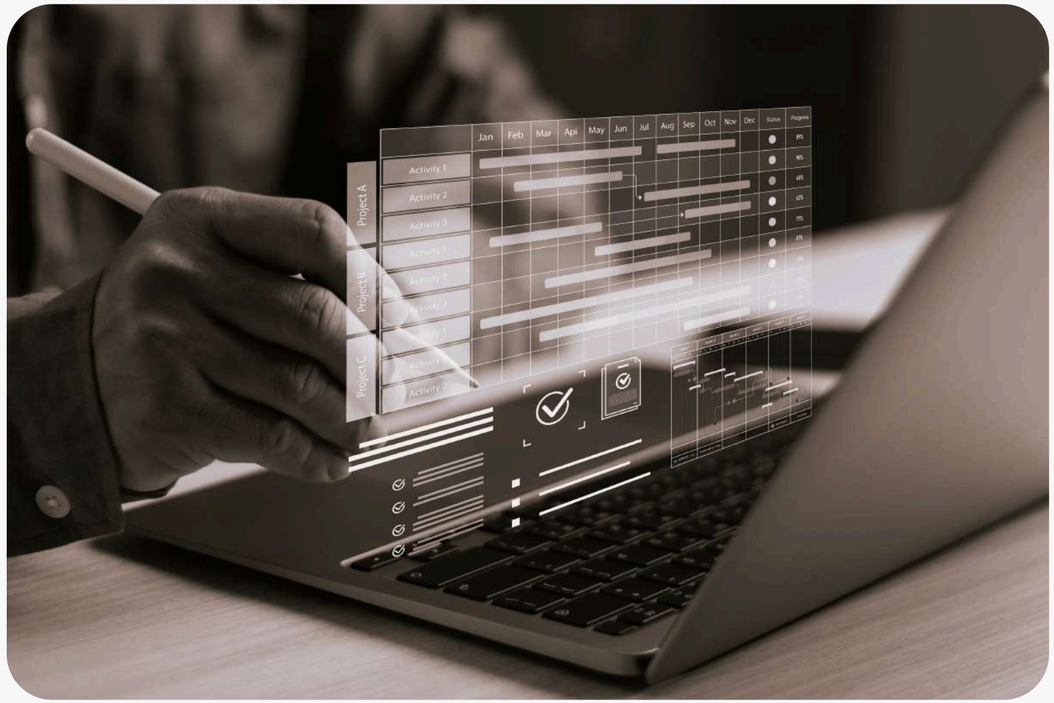
- Enables real-time monitoring of project health
- Facilitates early identification of **delays** and **risks**

## Enhancing Stakeholder Transparency

- Provides clients and stakeholders with up-to-date project insights
- Improves trust through accessible dashboards and reports

# Core Features and Roles

# Core Features for Impactful Monitoring




## Centralized Multi-Project Oversight

- Manage multiple projects and clients from a single platform for unified control.
- Assign roles and responsibilities to streamline collaboration across teams.

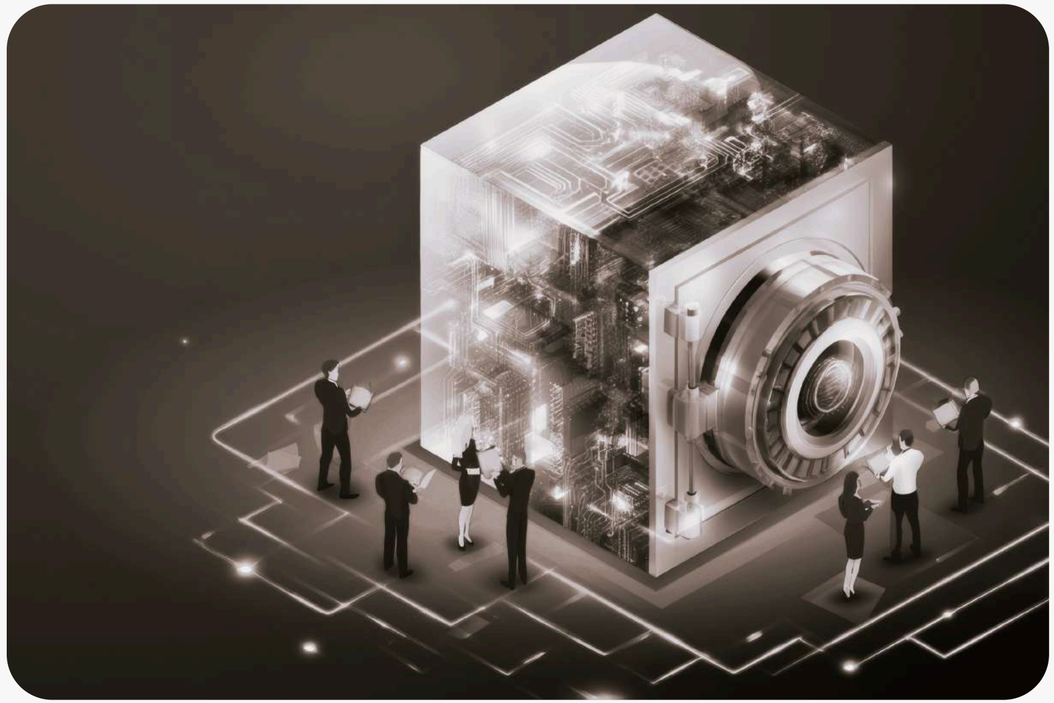
## Actionable Dashboards & Reporting

- Visualize project health with real-time dashboards showing **completion rates**, overdue items, and key metrics.
- Generate standard and custom reports for data-driven decision making.

## Mobile-First Field Updates

- Enable on-site staff to submit progress updates and log time directly from mobile devices.
- Capture and upload photos as attachments for instant project documentation.

# Role-Based Access for Security



### Granular Access Control

Distinct roles ensure only authorized users can access sensitive project data.

### Administrator Oversight

Admins configure system settings, manage user accounts, and assign roles.

### Project Manager Authority

Project managers create and manage projects, milestones, and team assignments.

### Empowered Team Members

Team members update task progress and log work hours directly in the system.

### Secure Collaboration

Role-based permissions protect data integrity while enabling efficient teamwork.

# Project and Task Management



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# Company and Client Management

### Centralized Company & Client Records

- Easily manage all organizations and clients in one place for streamlined project oversight.

### Project Grouping by Organization

- Associate each project with a specific company to enable focused tracking and reporting.

### High-Level Metrics at a Glance

- Instantly view **active projects**, **completion rates**, and **budget status** per company for rapid decision-making.

# Project Lifecycle Management

## Comprehensive Project Oversight

- Initiate, update, and finalize projects with a full **audit trail** for accountability
- Maintain project integrity through status changes: Planned, Active, On Hold, Completed, or Cancelled

## Team Assignment & Document Control

- Assign multiple team members with defined roles and allocation percentages
- Centralize key project documents and attachments for easy access and compliance

## Real-Time Progress Monitoring

- Track project **completion rates** and open/closed tasks at a glance
- Identify upcoming milestones and overdue items for proactive management

# Milestone Tracking for Control

### Structured Milestone Planning

- Set clear due dates and status for each milestone
- Ensure every milestone is measurable and time-bound

### Task Grouping for Clarity

- Organize related tasks under specific milestones
- Enhances visibility into project phases and dependencies

### Proactive Risk Identification

- Automatically flag overdue or at-risk milestones
- Enable early intervention to keep projects on track

# Task Management and Visualization

### Flexible Task Creation

- Define tasks with detailed descriptions, start/end dates, and estimated hours.
- Set task priorities and establish dependencies for accurate scheduling.

### Collaborative Assignments

- Assign multiple team members to each task, supporting diverse project roles.
- Track allocation percentages for balanced workload distribution.

### Visual Progress Tracking

- Utilize Kanban board views to monitor task status: To Do, In Progress, Blocked, Done.
- Filter and sort tasks by assignee, status, priority, or dates for focused oversight.

# Progress, Time, and Documents



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# Progress Updates and History

## Real-Time Progress Updates

- Enable team members to submit updates directly on tasks or projects for immediate visibility.

## Track Completion and Status

- Monitor **percent complete** and concise status summaries to assess project health at a glance.

## Historical Trends and Timelines

- Access a comprehensive timeline of updates to identify trends, bottlenecks, and improvement areas.

# Time Logging and Effort Tracking

## Effort Tracking for Accountability

Team members record hours spent on each assigned task, ensuring accurate tracking of project effort.

## Manager Oversight and Analysis

Managers access consolidated views of total hours by user, project, or time period for informed decision-making.

## Quality Control with Rule Checks

Optional approval workflows and rule enforcement (e.g., daily hour limits) maintain data integrity and compliance.

# Document and Attachment Management

### Centralized Document Storage

- Attach files directly to projects and tasks for streamlined access.

### Comprehensive Metadata Management

- Each file is tagged with uploader, date, and file details for **quick retrieval**.

### Seamless Access Across Platforms

- Preview or download documents from both web and mobile interfaces.

# Insights, Notifications, and Mobile

# Dashboards for Real-Time Insights

### Comprehensive Project Health at a Glance

- Visual indicators highlight **on-track, delayed, or at-risk** projects
- Quickly identify overdue tasks and upcoming milestones

### Dynamic Task and Milestone Analytics

- Interactive charts display task status distribution and milestone progress
- Monitor completion trends to spot bottlenecks early

### Organization-Wide Oversight

- Aggregate dashboards provide leadership with **real-time visibility** across all projects
- Enable data-driven decisions for resource allocation and risk management

# Automated Notifications and Alerts

## Real-Time Event Notifications

- Immediate alerts for task assignments, status changes, and project updates
- Supports in-app, email, and push channels for broad reach

## Proactive Overdue Alerts

- Automatic reminders for overdue tasks and milestones to drive timely action

## Personalized Notification Settings

- Users can configure preferences for notification types and frequency

# Mobile App for Field Teams

### Real-Time Task Management

- Access and update assigned tasks directly from the field

### Instant Time Logging & Photo Uploads

- Quickly log work hours and attach site photos as progress evidence

### Reliable Offline Functionality

- Continue updates even in remote areas with **offline-first** support

# Security, Performance, and Maintenance

# Robust Security and Compliance

## End-to-End Data Protection

- All API traffic secured with **HTTPS/TLS** encryption
- User passwords stored as **salted hashes**—never in plain text

## Granular Access Controls

- Role-based permissions restrict access to sensitive project data
- API endpoints enforce **authorization checks** to prevent privilege escalation

## Regulatory Compliance Ready

- Designed to support **data protection policies** and organizational IT security standards

# Performance and Scalability

## Handles High User Load

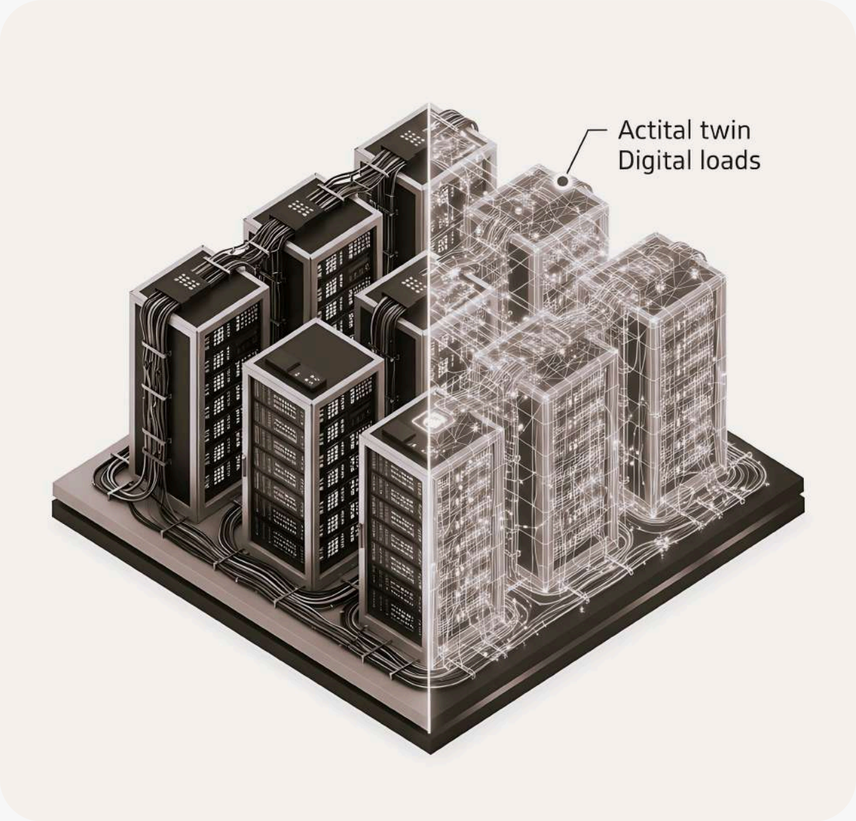
- Engineered to support **200+ concurrent users** without performance degradation

## Real-Time Data Access

- API responses optimized for **1-2 second** turnaround, enabling instant project insights

## Scalable System Architecture

- Stateless API services allow **horizontal scaling** to meet growing organizational needs



# Maintainability and Extensibility

### Layered Clean Architecture

- Separation of concerns enables rapid updates and clear code organization.
- Distinct Domain, Application, Infrastructure, and API layers for maintainability.

### Automated Testing Coverage

- Critical business logic is protected by **automated unit and integration tests**.
- Ensures system reliability during enhancements or refactoring.

### Modular & Extensible Design

- Feature-based structure in both web and mobile clients supports easy addition of new modules.
- Architecture supports future growth and integration with new technologies.

# Key Takeaways for Project Leaders

### Real-Time Project Visibility

Dashboards and reports provide instant insight into project health and progress.

### Centralized Data for Proactive Management

Unified tracking of tasks, milestones, and time logs enables early detection of risks and delays.

### Empowering Teams and Stakeholders

Web and mobile apps allow all roles to update, monitor, and collaborate efficiently.